

Problem

If $f(t) = e^{-2|t|}$, find $\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega$.

Step-by-step solution

Step 1 of 1

We have

$$f(t) = e^{-2|t|}$$

now,

$$J = \int_{-\infty}^{\infty} |f(\omega)|^2 d\omega = 2\pi \int_{-\infty}^{\infty} f^2(t) dt$$

$$= 2\pi \int_{-\infty}^{\infty} [e^{-2|t|}]^2 dt$$

$f(-t) = f(t)$ i.e. it is even function so

$$J = 4\pi \int_0^{\infty} e^{-4t} dt$$

$$= \frac{4\pi}{-4} [e^{-4t}]_0^{\infty}$$

$$= -\pi [e^{-\infty} - e^{-0}] = -\pi [0 - 1]$$

$$= \boxed{\pi}$$